

Current Computational Results for Lorentz/SEP Paper

September 14, 2026

This note collects the current generated computational fragments in the style of the latest manuscript draft. The raw CSV files have been refreshed so that every Lazy SEP row uses coordinate skew-skew cuts and reports coordinate skew-skew residuals. In the max-distance and TTRS tables, each SDP gap uses the feasible point recovered from that method’s own first column as its upper bound; neither Gurobi nor a beta reference supplies an SDP upper bound. Throughout the note, “tight” has the single definition that this method-specific relative gap is at most 10^{-5} . The larger generated panels also calculate relative rank-one residuals and recovered-point feasibility violations, but those diagnostics do not enter the tight count.

Displayed formulations. The following equations are included so the generated fragments compile outside `paper/Lorentz.tex`.

$$\min\{c^\top x + d^\top y + y^\top R x : x \in \mathcal{E}_x, y \in \mathcal{E}_y\}. \quad (1)$$

$$\min\{c^\top x + d^\top y + x^\top Q x + y^\top P y + y^\top R x : x \in \mathcal{E}_x, y \in \mathcal{E}_y\}. \quad (2)$$

$$\min\{\langle C, U \rangle : \text{tr } X \leq 1, \text{tr } Y \leq 1, U \succeq 0\}. \quad (3)$$

$$(3) \text{ with } Z = \mathcal{W}^*(T), \langle T, X \rangle = 0 \ \forall X \in \mathcal{A}(n) \otimes \mathcal{A}(m), T \succeq 0. \quad (4)$$

1 Pure Bilinear Instances

We generated 120 random instances of (1) with $\mathcal{E}_x = \{x \in \mathbb{R}^m : \|x\| \leq 1\}$ and $\mathcal{E}_y = \{y \in \mathbb{R}^n : \|y\| \leq 1\}$. For each candidate instance, $R \in \mathbb{R}^{n \times m}$ was sampled with independent standard normal entries. Let $u_1 \in \mathbb{R}^n$ and $v_1 \in \mathbb{R}^m$ be the leading left and right singular vectors of R . We then set $c = \alpha v_1 + \sigma g$ and $d = \alpha u_1 + \sigma h$, where g and h have independent standard normal entries, $\alpha = 3.0$, and $\sigma = 0.1$. Finally, the triple (c, d, R) was divided by $\max\{\|c\|, \|d\|\}$, so that $\max\{\|c\|, \|d\|\} = 1$. For each size we scanned seeds starting from zero and selected the first ten candidates for which the Shor relative gap $(z_{\text{exact}} - z_{\text{Shor}})/|z_{\text{Shor}}|$ exceeded 0.01. We compared the Shor SDP relaxation, Full SEP, Lazy SEP, the Dual LOP/TRS bisection method, and Gurobi. Gurobi was run after the three conic methods with explicit bounds $-1 \leq x_j \leq 1$, $-1 \leq y_i \leq 1$ for each i and j , the two ball constraints, and time limit $\max\{5, 3 t_{\text{max}}\}$, where t_{max} is the maximum time used by the three conic methods on that instance. Lazy SEP added at most 50 violated coordinate skew-skew equalities in each outer-approximation round, using violation tolerance 10^{-7} . The Dual LOP/TRS method used bisection tolerance 10^{-6} .

The median times required by different methods on these test instances are given in Table 1. In the table, the column $n \times m$ gives the dimensions of y and x , respectively. Ten cases were run for each size. The Shor, Full SEP, Lazy SEP, LOP/TRS, and Gurobi columns give median wall-clock

Table 1: Median wall-clock times for bilinear instances.

$n \times m$	cases	Shor	Full SEP	Lazy SEP	LOP/TRS	Gurobi	Gurobi opt
2×2	10	1 ms	1 ms	1 ms	18 ms	25 ms	10
4×4	10	1 ms	2 ms	5 ms	23 ms	1.81 s	9
6×6	10	1 ms	11 ms	21 ms	24 ms	5.00 s	0
8×8	10	2 ms	49 ms	82 ms	28 ms	5.00 s	0
10×10	10	2 ms	321 ms	318 ms	31 ms	13.29 s	0
15×15	10	3 ms	12.00 s	3.10 s	40 ms	100.67 s	0
20×20	10	5 ms	481.33 s	51.20 s	60 ms	1444.06 s	0
4×8	10	1 ms	7 ms	16 ms	27 ms	5.00 s	0
8×12	10	2 ms	248 ms	267 ms	33 ms	8.77 s	0
10×20	10	3 ms	7.14 s	2.24 s	51 ms	77.54 s	0
15×20	10	4 ms	59.18 s	15.22 s	54 ms	249.99 s	0
15×25	10	5 ms	249.54 s	24.95 s	65 ms	748.66 s	0

times for the five methods. The Gurobi opt column gives the number of instances solved to global optimality by Gurobi within its adaptive time limit.

In Table 2 we give additional numerical diagnostics for the same instances as in Table 1. By construction none of the Shor instances has a lower bound that agrees with the exact conic value within 10^{-5} . The “Shor max gap” column gives the largest gap between the exact conic value and the Shor lower bound. The “Conic max diff” column is the largest objective difference among all the exact conic methods over all instances of that size. The Lazy SEP columns give the median/maximum outer-approximation rounds, total added coordinate skew-skew cuts and the largest final coordinate skew-skew equation violation. The LOP/TRS columns report the median/maximum TRS separation oracle solves for the Dual LOP/TRS method as well as the maximum final bisection bracket width.

Table 2: Additional diagnostics for bilinear instances.

$n \times m$	Shor	Conic	Lazy SEP			LOP/TRS	
	max gap	max diff	rounds	cuts	max skew	oracles	max gap
2×2	$2.9 \cdot 10^{-1}$	$5.5 \cdot 10^{-8}$	2/2	1/1	$2.3 \cdot 10^{-14}$	23/23	$8.9 \cdot 10^{-7}$
4×4	$2.4 \cdot 10^{-1}$	$3.2 \cdot 10^{-8}$	2/2	36/36	$5.8 \cdot 10^{-12}$	23/23	$8.3 \cdot 10^{-7}$
6×6	$1.9 \cdot 10^{-1}$	$1.2 \cdot 10^{-7}$	3/4	100/150	$1.3 \cdot 10^{-8}$	23/23	$8.1 \cdot 10^{-7}$
8×8	$1.4 \cdot 10^{-1}$	$3.2 \cdot 10^{-8}$	5/6	200/250	$2.2 \cdot 10^{-9}$	23/23	$7.5 \cdot 10^{-7}$
10×10	$1.2 \cdot 10^{-1}$	$4.1 \cdot 10^{-8}$	9/12	400/550	$2.1 \cdot 10^{-9}$	23/23	$7.3 \cdot 10^{-7}$
15×15	$6.8 \cdot 10^{-2}$	$1.2 \cdot 10^{-8}$	22/37	1025/1800	$3.0 \cdot 10^{-9}$	23/23	$6.7 \cdot 10^{-7}$
20×20	$6.7 \cdot 10^{-2}$	$5.5 \cdot 10^{-9}$	63/76	3100/3750	$9.9 \cdot 10^{-9}$	23/23	$6.6 \cdot 10^{-7}$
4×8	$1.8 \cdot 10^{-1}$	$3.2 \cdot 10^{-7}$	3/4	100/150	$1.8 \cdot 10^{-9}$	23/23	$7.9 \cdot 10^{-7}$
8×12	$1.1 \cdot 10^{-1}$	$1.2 \cdot 10^{-7}$	9/11	400/500	$2.2 \cdot 10^{-9}$	23/23	$7.1 \cdot 10^{-7}$
10×20	$8.7 \cdot 10^{-2}$	$8.5 \cdot 10^{-9}$	20/30	950/1450	$2.2 \cdot 10^{-9}$	23/23	$6.8 \cdot 10^{-7}$
15×20	$6.0 \cdot 10^{-2}$	$2.3 \cdot 10^{-8}$	41/54	2000/2650	$1.9 \cdot 10^{-9}$	23/23	$6.5 \cdot 10^{-7}$
15×25	$5.5 \cdot 10^{-2}$	$3.1 \cdot 10^{-7}$	46/82	2250/4050	$2.3 \cdot 10^{-9}$	23/23	$6.5 \cdot 10^{-7}$

To summarize these results, Lazy SEP is slower than Full SEP on smaller instances due to the overhead of checking for violated skew-skew constraints and repeated conic solves, but the time for Full SEP blows up faster as problem size increases. The Dual LOP/TRS bisection algorithm is very fast and robust and scales much better than Full SEP or Lazy SEP. These problems are difficult for Gurobi, even with the explicit variable bounds added, and cannot be solved in time competitive to the conic methods.

2 Max-Distance Instances

We generated 60 random max-distance instances, that is, instances of (2) of the form

$$\max\{\|(Ax + a) - (By + b)\|^2 : \|x\| \leq 1, \|y\| \leq 1\},$$

with $n = m$, so that $\mathcal{E}_x = \{Ax + a : \|x\| \leq 1\}$ and $\mathcal{E}_y = \{By + b : \|y\| \leq 1\}$ are the two ellipsoids whose farthest pair of points is sought. The maps A and B were generated as $U\Sigma V^\top$ with Haar-random orthogonal U, V and singular values log-uniform on $[1, 5.0]$, and the centers a and b have independent normal entries. Because the Shor relaxation (3) is already exact on most max-distance instances, the instances reported here were selected by a rank-only screen, applied to seeds starting from zero. The screen first solves (3) and discards the instance when the two largest eigenvalues of the optimal U satisfy $\lambda_1/\lambda_2 > 10^4$, treating a numerically negative λ_2 by using $|\lambda_2|$ in the denominator. Such a solution is effectively rank one, in which case U is itself a feasible point attaining the bound and the relaxation is exact. For the remainder, the screen re-solves the Shor relaxation with the Shor objective held fixed up to solver tolerance and with a random linear objective; this exposes rank-one optima when the original solve returned a higher-rank point on a flat optimal face. If this second solve is still not effectively rank one, the instance is retained; no feasible reference value is computed during this preselection step. Such instances are rare: retaining 10 per size required scanning up to 16970 seeds. The reported results are therefore conditional on the Shor relaxation not being certified rank one by this screen, and are not representative of the class as a whole. We compared the Shor SDP relaxation (3), Shor strengthened by the KRON constraint $K(Z) \succeq 0$, Lazy KRON, Full SEP (4), and Lazy SEP. Unlike the bilinear problem of Section 3, none of these relaxations is guaranteed exact for the class (2). For each SDP method, we extracted (x, y) from the first column of its optimal moment matrix, verified the two ball constraints to tolerance 10^{-7} , and evaluated the original minimization objective at that point. This gives a method-specific feasible upper bound; no Gurobi incumbent or point recovered by another method is used to calculate an SDP gap. The retained Gurobi runs used explicit bounds $-1 \leq x_j \leq 1$, $-1 \leq y_i \leq 1$, the two ball constraints, and time limit $\max\{5, 3t_{\max}\}$, where $t_{\max}$ is the maximum conic-method time available when that Gurobi run was started. These Gurobi rows were not rerun after the conic refresh and are reported only as an independent comparison. Lazy KRON added at most 50 violated KRON cuts in each outer-approximation round. Lazy SEP added at most 50 violated coordinate skew-skew equalities in each round. Both used relative violation tolerance 10^{-7} .

Table 3: Median wall-clock times for max-distance instances.

n	cases	Shor	KRON	Lazy KRON	Full SEP	Lazy SEP	Gurobi	Gurobi opt
2	10	1 ms	3 ms	4 ms	3 ms	3 ms	55 ms	10
4	10	1 ms	10 ms	8 ms	7 ms	7 ms	4.93 s	5
6	10	2 ms	69 ms	10 ms	21 ms	14 ms	5.00 s	0
8	10	2 ms	605 ms	22 ms	84 ms	32 ms	5.00 s	0
10	10	2 ms	4.60 s	38 ms	398 ms	58 ms	13.81 s	0
15	10	4 ms	611.65 s	151 ms	16.33 s	251 ms	1543.72 s	0

The median times required by the different methods are given in Table 3. In the table, the column n gives the dimensions of y and x . The Gurobi opt column gives the number of instances solved to global optimality by Gurobi within its adaptive time limit. The Full KRON solve for $n = 15$, seed 278, used two MOSEK threads after repeated memory-related termination with the default parallel setting; its formulation and tolerances were unchanged.

Table 4 reports bound quality on the same instances. For Shor, Full KRON, and Full SEP (labeled SEP), the tight column counts instances whose method-specific relative recovered-point

gap is at most 10^{-5} , while max gap gives the largest such gap over the instances of that size. If z_{sdp} is the SDP lower bound and z_{rec} is that method’s recovered feasible upper bound, this gap is $(z_{\text{rec}} - z_{\text{sdp}})/\max\{1, |z_{\text{rec}}|, |z_{\text{sdp}}|\}$. Lazy KRON matched the Full KRON tight count at all other sizes, but at $n = 2$, 9/10 versus 10/10 (Lazy KRON max gap $1.7 \cdot 10^{-5}$); at $n = 4$, 10/10 versus 9/10 (Lazy KRON max gap $8.2 \cdot 10^{-6}$); at $n = 8$, 8/10 versus 9/10 (Lazy KRON max gap $2.3 \cdot 10^{-1}$); at $n = 10$, 9/10 versus 10/10 (Lazy KRON max gap $2.5 \cdot 10^{-5}$); at $n = 15$, 8/10 versus 9/10 (Lazy KRON max gap $1.4 \cdot 10^{-1}$). Lazy SEP matched the Full SEP tight count at every size. The Lazy KRON and Lazy SEP columns give the median/maximum outer-approximation rounds and total added cuts. The max viol. column is the largest final relative KRON cut violation, while max skew is the largest final relative coordinate skew-skew equation violation.

Table 4: Bound quality and diagnostics for max-distance instances.

n	Shor		KRON		SEP		Lazy KRON			Lazy SEP		
	tight	max gap	tight	max gap	tight	max gap	rounds	cuts	max viol.	rounds	cuts	max skew
2	0	$8.8 \cdot 10^{-1}$	10	$1.3 \cdot 10^{-6}$	10	$2.7 \cdot 10^{-6}$	5/5	4/4	$8.3 \cdot 10^{-8}$	1/1	0/0	$1.5 \cdot 10^{-8}$
4	0	$7.7 \cdot 10^{-1}$	9	$2.2 \cdot 10^{-5}$	10	$7.3 \cdot 10^{-7}$	6/9	5/8	$6.1 \cdot 10^{-8}$	1/1	0/0	$5.9 \cdot 10^{-8}$
6	0	$7.6 \cdot 10^{-1}$	10	$2.4 \cdot 10^{-6}$	10	$3.3 \cdot 10^{-8}$	4/7	4/6	$8.4 \cdot 10^{-8}$	1/2	0/50	$2.6 \cdot 10^{-9}$
8	0	$7.3 \cdot 10^{-1}$	9	$2.3 \cdot 10^{-1}$	10	$5.2 \cdot 10^{-8}$	6/24	6/23	$10.0 \cdot 10^{-8}$	1/2	0/50	$1.1 \cdot 10^{-8}$
10	0	$6.5 \cdot 10^{-1}$	10	$1.0 \cdot 10^{-6}$	10	$8.4 \cdot 10^{-8}$	7/12	6/11	$9.6 \cdot 10^{-8}$	1/2	0/50	$2.0 \cdot 10^{-9}$
15	0	$5.5 \cdot 10^{-1}$	9	$1.4 \cdot 10^{-1}$	10	$1.3 \cdot 10^{-7}$	8/20	7/19	$9.0 \cdot 10^{-8}$	1/4	0/150	$1.0 \cdot 10^{-8}$

3 Ball-Constraints TTRS Instances

We ran Lazy SEP on the 212 centered diagonal TTRS instances from the ball-constraints repository, and Full SEP on 212 instances: all instances with $n \in \{5, 10, 20\}$. Shor and KRON are omitted from this panel. The implementation uses the transpose of the paper’s bilinear convention. It forms the square SEP lift

$$Z = \begin{pmatrix} 1 & y^\top \\ x & V \end{pmatrix}, \quad y = \text{diag}(a)x + b,$$

and represents $V_{ij} = x_i y_j = x_i(a_j x_j + b_j)$ linearly through the x -moment matrix. Full SEP imposes all skew-skew equations at once, while Lazy SEP adds violated coordinate skew-skew equations by separation. For every SDP row, bound quality is measured using that method’s own recovered first-column feasible point as an upper bound; neither Gurobi nor the corrected beta values are used to calculate an SDP gap.

Table 5: Results on TTRS instances from [?]

n	cases	Full SEP			Lazy SEP			Gurobi		
		time	tight	max gap	time	tight	max gap	time	tight	max gap
5	38	13 ms	38	$4.9 \cdot 10^{-7}$	25 ms	38	$1.2 \cdot 10^{-6}$	49 ms	38	$8.7 \cdot 10^{-7}$
10	70	457 ms	70	$2.3 \cdot 10^{-7}$	144 ms	70	$2.0 \cdot 10^{-7}$	447 ms	69	$1.5 \cdot 10^{-4}$
20	104	1016.15 s	104	$3.5 \cdot 10^{-7}$	2.18 s	104	$7.5 \cdot 10^{-7}$	487.45 s	93	$1.5 \cdot 10^{-4}$

4 Larger Generated Instances

The displayed labels match the current manuscript labels only by reference. The blocks below are intended as add-on fragments, not automatic replacements for the existing tables.

Table 6: Additional diagnostics for Lazy SEP on TTRS instances from [?]

n	rounds	cuts	max skew
5	2/2	80/100	$7.8 \cdot 10^{-9}$
10	2/2	100/100	$8.4 \cdot 10^{-9}$
20	2/3	100/200	$3.9 \cdot 10^{-8}$

Larger max-distance instances

We generated 30 additional max-distance instances using the same generator and rank-one Shor prefilter as in Tables 3 and 4. Full KRON, Full SEP, and Gurobi were omitted. The table columns therefore retain only the large-scale methods, Lazy KRON and Lazy SEP. In the diagnostics table, tight means that the relative gap between the relaxation bound and the objective value of the recovered first-column point is at most 10^{-5} . Relative rank-one residuals and recovered-point feasibility violations are also calculated and retained in the summary CSV, but neither enters the definition of tight.

Table 7: Results for larger max-distance instances

n	cases	Lazy KRON			Lazy SEP			Lazy KRON			Lazy SEP		
		time	tight	max gap	time	tight	max gap	rounds	cuts	max viol.	rounds	cuts	max skew
20	10	294 ms	8	$1.7 \cdot 10^{-5}$	1.01 s	10	$1.7 \cdot 10^{-7}$	4/9	3/8	$4.7 \cdot 10^{-8}$	1/2	0/50	$2.8 \cdot 10^{-8}$
25	10	3.85 s	9	$9.2 \cdot 10^{-5}$	3.20 s	10	$9.5 \cdot 10^{-8}$	3/7	2/6	$6.7 \cdot 10^{-8}$	1/1	0/0	$1.6 \cdot 10^{-9}$
30	10	14.54 s	9	$1.8 \cdot 10^{-5}$	9.17 s	10	$1.7 \cdot 10^{-8}$	4/7	3/6	$8.5 \cdot 10^{-8}$	1/3	0/100	$1.5 \cdot 10^{-9}$

Larger generated TTRS instances

We generated 30 centered affine-diagonal TTRS instances and used Shor only as a prefilter to discard instances already certified by rank-one recovery. The reported large-scale methods are Lazy KRON and Lazy SEP. Since no beta or Gurobi reference is used in this generated panel, tight means that the relative gap between a relaxation bound and the objective value of its recovered point is at most 10^{-5} . Relative rank-one residuals and recovered-point feasibility violations are also calculated and retained in the summary CSV, but neither enters the definition of tight.

Table 8: Results for additional TTRS instances

n	cases	Lazy KRON			Lazy SEP			Lazy KRON			Lazy SEP		
		time	tight	max gap	time	tight	max gap	rounds	cuts	max viol.	rounds	cuts	max skew
20	10	309 ms	4	$4.6 \cdot 10^{-5}$	1.01 s	10	$1.6 \cdot 10^{-7}$	4/18	4/17	$8.8 \cdot 10^{-8}$	1/5	0/100	$7.5 \cdot 10^{-8}$
25	10	1.47 s	6	$3.8 \cdot 10^{-1}$	3.69 s	10	$3.8 \cdot 10^{-6}$	7/22	6/21	$7.8 \cdot 10^{-8}$	1/10	0/900	$9.9 \cdot 10^{-8}$
30	10	20.14 s	4	$4.2 \cdot 10^{-1}$	12.15 s	10	$5.4 \cdot 10^{-7}$	6/37	5/36	$9.6 \cdot 10^{-8}$	2/6	52/500	$9.2 \cdot 10^{-8}$

5 Noxious-Facility Instances

Table 9: Distance upper bounds for regular m -gons

m	Shor	RLT	KRON	Full SEP
3	1.414214	1.414214	1.174750	1.114373
4	1.414214	1.414214	1.133203	1.001735
5	1.414214	1.414214	1.163184	1.025778
6	1.414214	1.414214	1.133203	1.001735
7	1.414214	1.414214	1.149584	1.010315
8	1.414214	1.414214	1.133203	1.001735
10	1.414214	1.414214	1.133203	1.001735
12	1.414214	1.414214	1.133203	1.001735
16	1.414214	1.414214	1.133203	1.001735

Table 10: Results for normalized random instances

m	cases	Shor gap median/max	RLT gap median/max	KRON gap median/max	SEP gap median/max
4	10	0.462/0.525	0.454/0.525	0.152/0.234	0.084/0.219
6	10	0.445/0.495	0.426/0.494	0.203/0.264	0.144/0.257
8	10	0.480/0.506	0.472/0.505	0.268/0.355	0.217/0.351
10	10	0.526/0.618	0.526/0.607	0.318/0.389	0.249/0.332
15	10	0.537/0.624	0.537/0.624	0.358/0.402	0.315/0.387
20	10	0.581/0.613	0.581/0.613	0.402/0.430	0.365/0.402

Table 11: Bounds for instance with points (??)

method	distance upper bound
Shor	1.2174074784
RLT	1.0382899391
KRON	0.8379375081
Full SEP	0.7257975088
exact value	0.7256779914